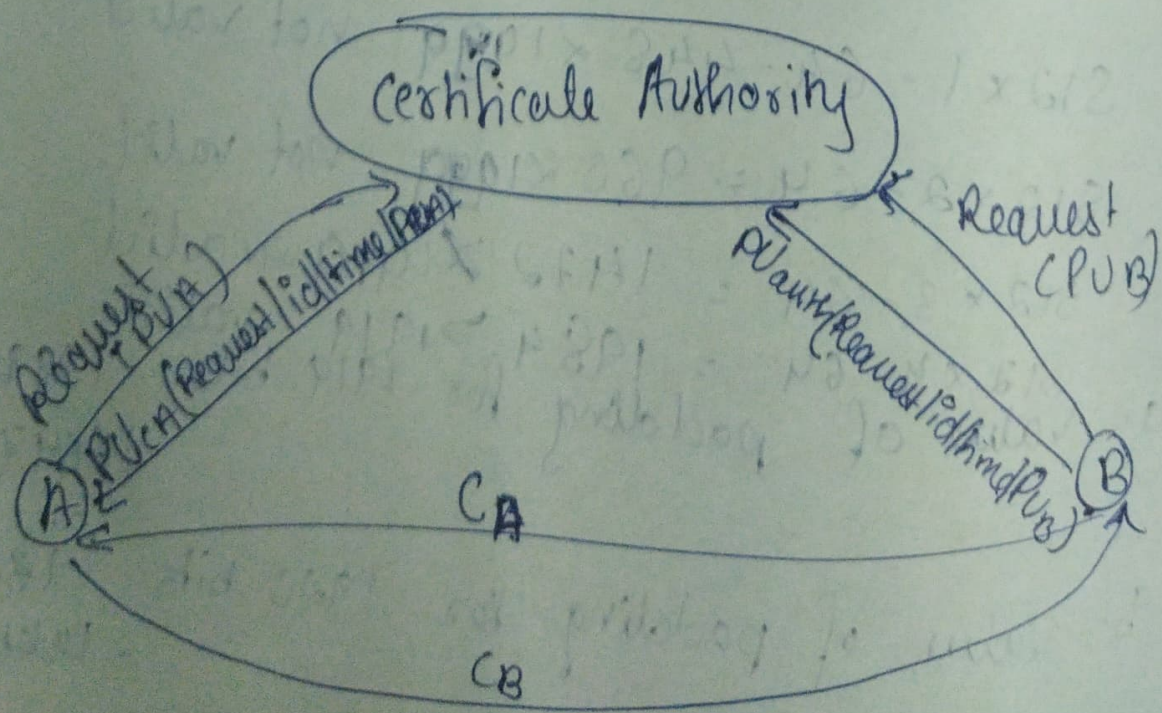
3) The certificate Authority is responsible for issuing digital certificate which ensures the authentication of the sender and receiver. It also ensures non-repudiation (denial of source of message) and integrity of the message.

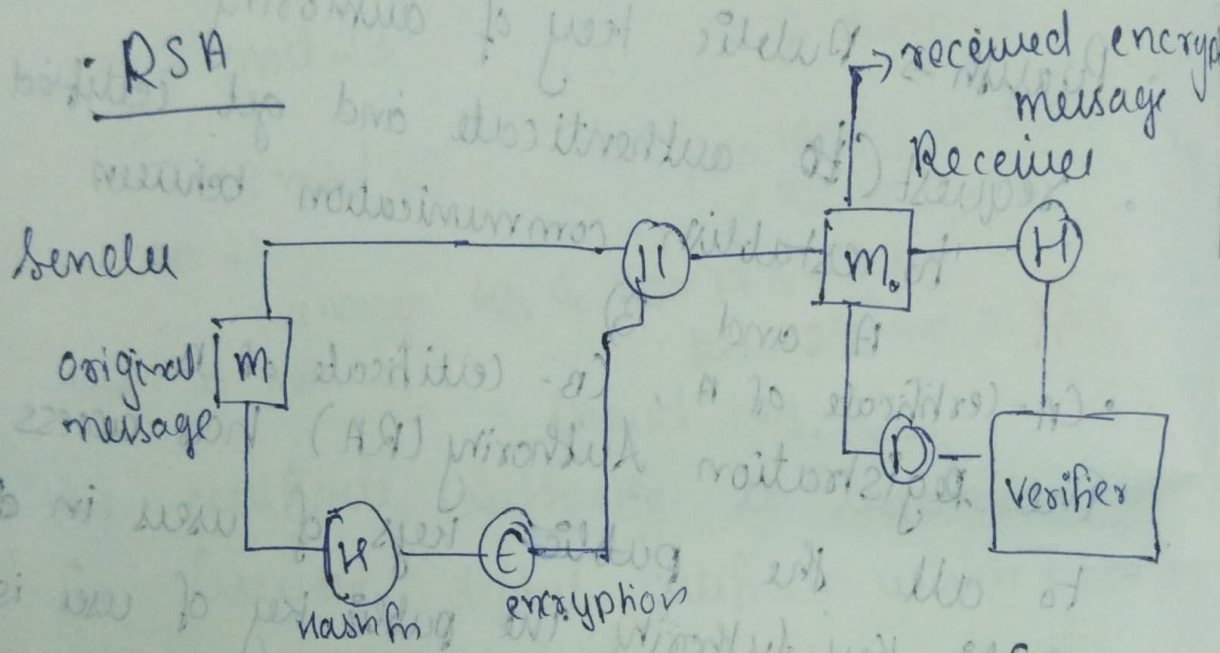


- PU_A → Public key of A
- PU_B → Public key of B
- PU_{auth} → Public key of authority
- request (to authenticate and get certified to establish communication between A and B)
- C_A - Certificate of A, C_B - Certificate of B

The Registration Authority (RA) has access to all the public keys of users in the Public-Key Authority. The public key of user is needed to identify the users between which communication needs to be made. This public key is sent to the Certificate Authority to validate the user. The Certificate Authority sends the digital certificate which has the sign of CA, date of issue, authenticated sender details which is send along with the message encrypted using sender's private key. This is then decrypted by the receiver using the public key of sender which is present in the Public-Key Authority.

4) For alice to be sure that the message is not altered, integrity of the message needs to be ensured one of the techniques for this is

- RSA



In this method, the message from the sender is hashed (message digest is made by padding using MD5 or SHA algorithm) which is then encrypted using the private key of the sender. This message is received by the receiver and hash fn is applied. The same message is decrypted using the public key of sender. Both these outputs are verified using verification function and if both outputs are same, the message is unaltered

6) The 4 phases of a virus are.

- dormant Phase
- Propogation Phase
- Trigger Phase
- Execution Phase

a) dormant Phase: during the dormant phase, the computer would be affected by the virus but it is idle; has not started propogation or any malicious activity. It is hence tough to detect the presence of virus during the dormant phase. Virus in this phase is limited to the location it was installed at.

b) Propogation phase: during this phase, the virus spreads across the system and gets into the files and folders present in the system. The virus becomes detectable at this phase.

c) Trigger Phase: Most often when a virus is installed by an external intruder, a trigger condition would be set to start the action of the virus. This can be a particular date, time or action such as download or redirect from some malicious sources

d) Execution Phase: The last phase of a virus before its executed and removed from the system is the execution phase. Once the virus is triggered, the execution phase begins and the virus corrupts the entire system or the area to be intended with the aim of data theft, forgery, financial attack, spying or even cluttering.